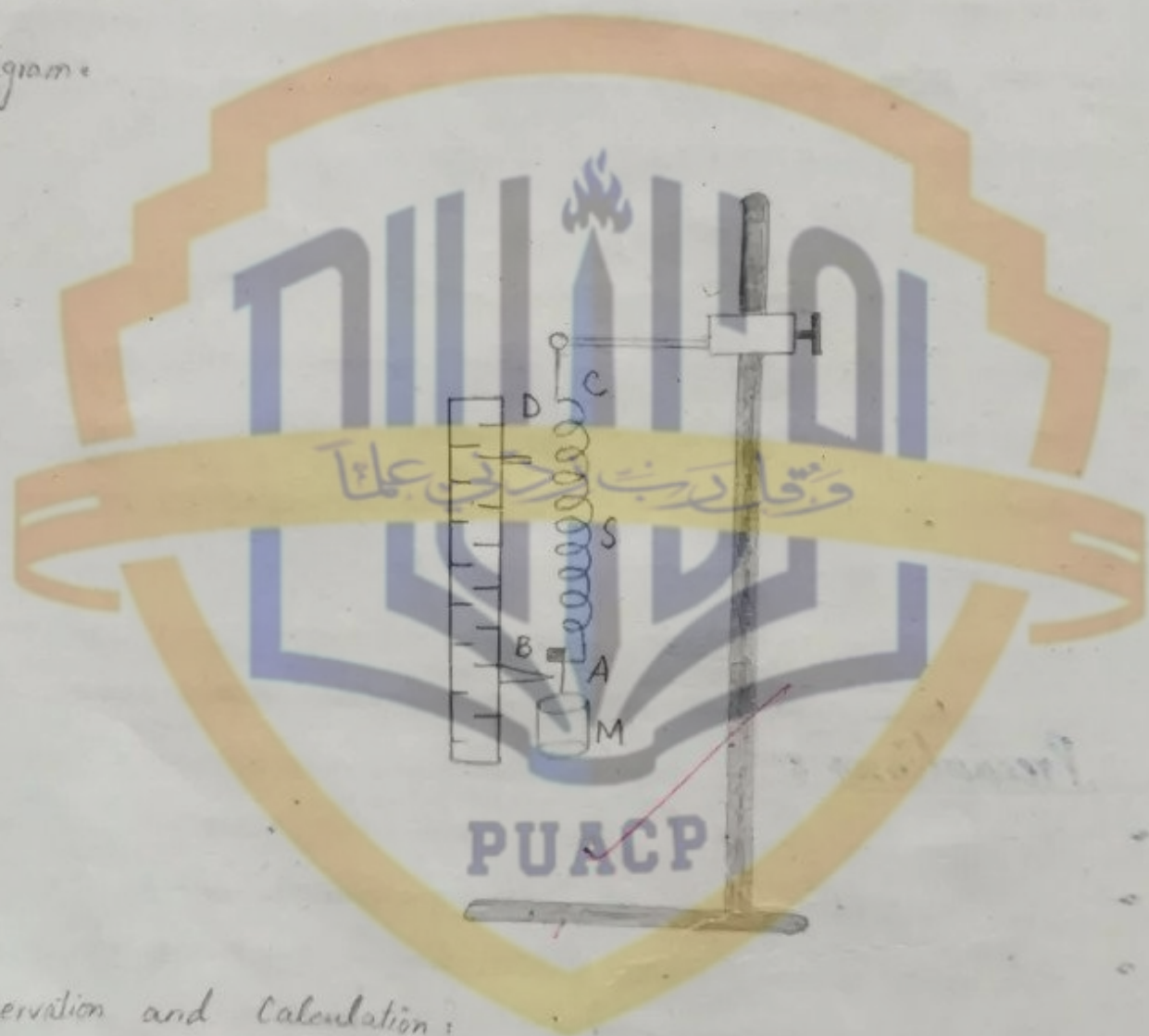


Object: To determine the elastic constant i.e. Modulus of rigidity of a wire by a spiral spring.

Apparatus:

- Spiral Spring
- heavy retort stand
- stop watch
- screw gauge
- Vernier calliper
- hanger with slotted weight

Diagram:



Observation and Calculation:

LC of Screw gauge = 0.01 mm = 0.001 cm
Zero Correction = Null cm

Corrected diameter of wire = i) 2.04 mm ii) 2.05 mm
iii) 2.03 mm
Mean diameter = $d = 2.04$ mm = 0.204 cm

Radius of wire = $r = \frac{d}{2} = 0.1021$ cm

Experiment Name: To determine the elastic constant i.e. Modulus of rigidity of a wire by a spiral spring

Apparatus :-

- Spiral spring
- heavy retort stand
- stopwatch
- Screw gauge
- Vernier calliper
- hanger with slotted weight

Procedure :-

- Set the apparatus. Take a flat spiral spring 'S' of external radius ' R_1 ' formed by closely winding of radius ' r ' on a wooden cylinder of uniform diameter.
- Find out diameter of wire with screw gauge and diameter of spiral spring with Vernier calliper.
- Fix one end of spring firmly in a heavy retort stand and suspend a hanger on the lower end of the spring.
- Calculate radius ' r ' and ' R ' and find out the effective radius $R = (R_1 - r)$ of the spiral spring.
- Pull the hanger gently downwards and then release it. It starts vibrating up and down.
- Find out the time for 20 vibrations with the help of stop watch and calculate the time period ' T '.
- Repeat this process several times by increasing load in the hanger.

L.C of Vernier Calliper = 0.1 cm

Corrected diameter of spiral spring = i) 2.9 cm
 ii) 2.95 cm iii) 2.96 cm

Mean diameter = $D = 2.93$ cm

Radius of spiral spring = $\frac{D}{2} = 1.47$ cm

Effective radius of spiral spring = $R = (R_1 - r) = 1.37$ cm

No. of turns of spiral spring = 200

Mass of spiral spring = $m = 125$ g.m

no. of obs.	Mass Suspended M g.m	Time for 20 vibrations				Time Period		Modulus of rigidity $\eta = \frac{16\pi^2 R^3 N}{8^4 T^2}$ (in dynes/cm ²)
		1	2	3	Mean time	$T = t/20$	T^2	
		Sec	Sec	Sec	t Sec	Sec	sec ²	
1	150	11.6	11.7	11.8	11.7	0.585	0.3422	6.31×10^{10}
2	200	12.0	12.1	12.2	12.1	0.625	0.390	6.9×10^{10}
3	250	13.5	13.4	13.6	13.5	0.69	0.462	7.1×10^{10}
4	300	14.2	14.3	14.4	14.3	0.715	0.511	7.5×10^{10}
5	350	14.9	14.8	15.0	14.9	0.745	0.556	7.9×10^{10}

$$\eta = \frac{16\pi^2 R^3 N}{8^4 T^2} \cdot \frac{(M+m)g}{T^2} = \frac{(16)(3.14)^2 (1.37)^3 (200)}{(0.1021)^4} \cdot \frac{350 + 125/3}{0.556}$$

$$\eta = 1125387267.704436$$

$$\eta = 7.9 \times 10^{10}$$

$$\text{Mean } \eta = \frac{6.3 \times 10^{10} + 6.9 \times 10^{10} + 7.1 \times 10^{10} + 7.5 \times 10^{10} + 7.9 \times 10^{10}}{5} = 7.14 \times 10^{10} \text{ dynes/cm}^2$$

Result:- Modulus of rigidity of wire = 7.14×10^{10} - dynes/cm²

Actual Value = 8.0×10^{10} dynes/cm²

$$\text{Percentage error} = \frac{\text{Actual value} - \text{calculated value}}{\text{Actual Value}} \times 100$$

$$= \frac{8.0 \times 10^{10} - 7.14 \times 10^{10}}{8.0 \times 10^{10}} \times 100$$

$$= 10.75\%$$

- Count the total number of turns N of the spring.
- Weigh the spring accurately and calculate its mass. let its be m grams.
- Calculate the modulus of rigidity of a wire by using;

$$\eta = \frac{16\pi^2 R^3 N}{h^4} \cdot \frac{M + m/3}{T^2}$$
- Repeat the experiment with different masses added in the hanger.

PUACP

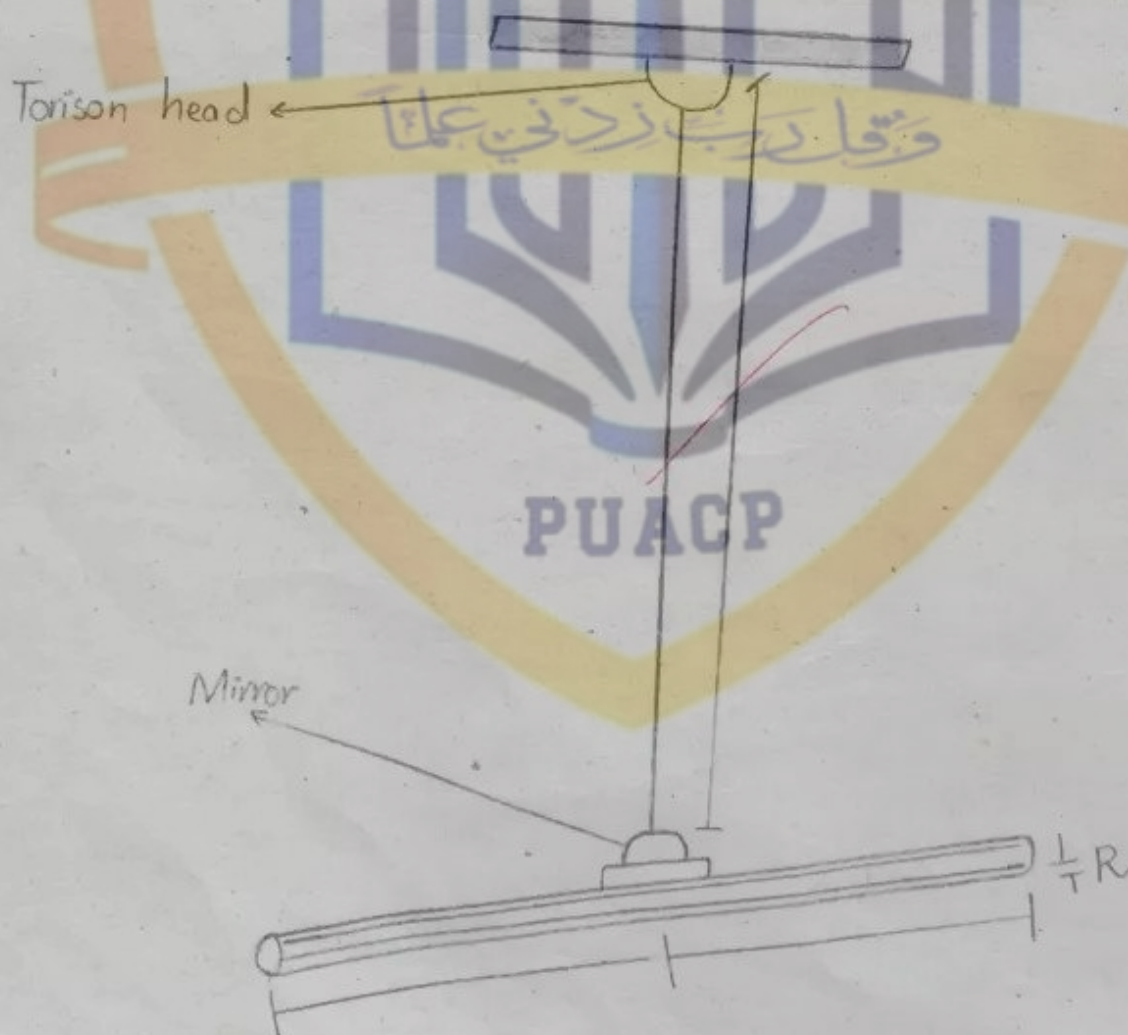
Experiment = To determine the modulus of rigidity of a wire

Oscillating rod (Dynamic method)

Apparatus =

- An Oscillating rod of some metallic solid cylinder fitted with a mirror in the centre
- A rigid support with torsion head
- Telescope
- Meter rod
- Vernier callipers
- Screw gauge
- Stop watch
- Wire of suitable length and radius
- Spring balanced

Diagram =



To determine the modulus of rigidity of a wire by Oscillating rod (Dynamic method)

APPARATUS:-

- ◆ An oscillating rod of some metallic solid cylinder fitted with a mirror in the centre
- ◆ A rigid support with torsion head
- ◆ Telescope
- ◆ Meter rod
- ◆ Vernier callipers
- ◆ Screw gauge
- ◆ stop watch
- ◆ Wire of suitable length and radius
- ◆ Spring balance

PROCEDURE:-

- ◆ Suspend an oscillating metallic rod horizontally from one end of given wire and other end is clamped in a rigid support.
- ◆ Place a knitting needle vertically in front of a small mirror at distance of 5cm away from its face.
- ◆ Adjust the eye piece of telescope to see the cross wire clearly. Focus the telescope from a distance of about 2 meters on the image of needle tip as the image is removed.
- ◆ In order to control the motion, tie one end of thread to the rod and other end to fixed point or support.
- ◆ Burn the thread by candle near the rod to make it vibrate in horizontal position.
- ◆ Start the stop watch when the needle just passes through the vertical cross-wire of telescope to right then count zero.

Observation and Calculation :-

- i- length of suspension wire ----- 96 ----- cm
 L.C of screw guage ----- 1.08 ----- mm ----- 0.04 ----- cm
 Zero error of screw guage ----- Null ----- mm ----- Null ----- cm
 Measure diameter of wire i) ----- 0.8 ----- cm ii) ----- 0.81 ----- cm iii) ----- 0.82 ----- cm
 Mean diameter of wire ----- 0.81 ----- cm
 Radius of wire = $r = d/2$ ----- 0.405 ----- cm
 ii- length of oscillating rod ----- 31 ----- cm
 L.C of Vernier calliper ----- 0.1 ----- cm
 Zero error of Vernier Calliper ----- 0.001 ----- cm
 Measure diameter of rod i) ----- 0.11 ----- cm ii) ----- 0.112 ----- cm iii) ----- 0.110 ----- cm
 Mean diameter of rod ----- 0.112 ----- cm , Radius of rod = R ----- 0.056 ----- cm
 Mass of Oscillating rod ----- 750 ----- grams
 Moment of inertia of cylindrical rod = $I = M \left(\frac{L^2}{12} + \frac{R^2}{4} \right) = 60068.34 \text{ gm-cm}^2$

No. of obs.	Time for 20 seconds				Time period $T = \frac{t}{20}$	T^2
	1 sec	2 sec	3 sec	Mean time t seconds		
1.	90	90.2	90.3	90.3	4.5	20.25
2.	100.1	100.2	100.3	100.2	5.1	26.1

$$\text{Modulus of rigidity} = \eta = \frac{8\pi/I}{4T^2} = \frac{8(3.14)(96)(60068.34)}{(0.405)^4 (4.8)}$$

Result =

The modulus of rigidity of the material of wire = $\eta = 7.0 \times 10^{10}$ dynes-cm²
 Actual value of η = 6.3×10^{10} dynes-cm²
 Percentage error = 10 %

- Measure the length of oscillating rod and length of suspension wire by a meter rod.
- Measure diameter of rod and Diameter of wire at different points with the help of vernier calliper.
- Measure mass of rod and calculate moment of inertia;

$$I = M \left(\frac{L^2}{12} + \frac{R^2}{4} \right) = \dots \dots \dots \text{gmm}^2$$

- With record observation, Modulus of rigidity;

$$\eta = \frac{8\pi l}{14T^2} \frac{I}{\dots}$$

Precautions :-

- The oscillating rod should remain in horizontal plane and should not vibrate up and down.
- The rod should vibrate equally on both sides of needle.
- The suspension wire should be free from kinks.
- The amplitude of vibration should be small to keep the wire within elastic limit.
- The diameter of wire should be determine very carefully at different points of suspension.
- Protect the apparatus from all disturbances during movement of time period.